INSTRUCTIONS:

1. **Write the table on the ruled side.**
2. **Write the reactions for all the confirmatory tests in each experiment on the unruled side.**

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 1

A] PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS: **DETECTION OF GROUP I ACID RADICALS**		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.
DETECTION OF GROUP II ACID RADICALS:		
Salt + conc. H_2SO_4 in a dry test tube. Above solution + Copper turnings. Heated strongly.	No reaction No reaction	Group II acid radical is absent.
DETECTION OF GROUP III ACID RADICAL(SO_4^{2-}):		
Clear solution of the salt in dil.HCl+ Barium chloride solution.	A white precipitate is formed. It is insoluble in excess of dil.HCl	III group acid radical is present Sulphate is present
$SO_4^{-2} + BaCl_2 \rightarrow \underset{\text{white ppt.}}{BaSO_4 \downarrow} + 2Cl^-$		
CONFIRMATORY TEST FOR SULPHATE (SO_4^{-2})		
Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution.	A white precipitate soluble in ammonium acetate solution	Sulphate is confirmed.
$SO_4^{-2} + Pb(CH_3COO)_2 \rightarrow \underset{\text{white ppt.}}{PbSO_4 \downarrow} + 2CH_3COO^-$		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.

DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original Solution + NH_4Cl solid +NH_4OH solution in excess + H_2S	No precipitate.	IV group basic radical absent.
DETECTION OF GROUP V BASIC RADICALS		
Original solution + NH_4Cl solid +NH_4OH solution in excess+$(NH_4)_2CO_3$ solution.	No precipitate.	V group basic radical absent.
DETECTION OF GROUP VI BASIC RADICALS		
Original solution + NH_4Cl solid + NH_4OH solution + ammonium phosphate solution.	White crystalline precipitate	VI group basic radical present. Mg^{2+} is present.
CONFIRMATORY TEST FOR MAGNESIUM (Mg^{+2})		
Original solution + NaOH solution.	White precipitate insoluble in NaOH solution	Mg^{2+} is confirmed.
$Mg^{2+} + (NH_4)_2HPO_4 \rightarrow Mg(NH_4)PO_4\downarrow$ $Mg^{2+} + 2NaOH \rightarrow Mg(OH)_2\downarrow + 2Na^+$		

RESULT:

The given salt contains __sulphate(SO_4^{-2})__ as acid radical and _____magnesium_(Mg^{+2})____ as basic radical.
Hence the given salt is $MgSO_4$ – Magnesium sulphate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 2

A] PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H_2SO_4 in a dry test tube.	Colourless gas is evolved. It gives dense white fumes with glass rod dipped in NH_4OH.	Group II acid radical is present. Cl- may be present.

CONFIRMATORY TEST FOR CHLORIDE (Cl^-)

1.Silver Nitrate test: Clear salt solution in dilute Nitric acid + few drops of silver nitrate solution.	A curdy white precipitate soluble in excess of ammonium hydroxide	Chloride is confirmed.

$$Cl^- + AgNO_3 \rightarrow NO_3^- + \underset{\text{white ppt.}}{AgCl\downarrow}$$

$$AgCl + 2NH_4OH \rightarrow \underset{\text{soluble complex}}{[Ag(NH_3)_2]Cl} + 2H_2O$$

2.Chromyl Chloride test: A pinch of salt + potassium dichromate crystals + few drops of conc.H_2SO_4. Pass the above vapours into NaOH solution. To the yellow solution add acetic acid and lead acetate solution.	Reddish brown vapours of Chromyl Chloride are evolved. Yellow solution. Yellow precipitate.	Chloride is confirmed.

$$2Cl^- + K_2Cr_2O_7 + 3H_2SO_4 \rightarrow SO_4^{-2} + K_2SO_4 + \underset{\text{chromyl chloride}}{2CrO_2Cl_2} + 3H_2O$$

$$CrO_2Cl_2 + 4NaOH \rightarrow Na_2CrO_4 + 2H_2O + 2NaCl$$

$$Na_2CrO_4 + Pb(CH_3COO)_2 \rightarrow \underset{\text{Yellow ppt.}}{PbCrO_4\downarrow} + 2CH_3COONa$$

C] ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	A pungent smell of ammonia which gives dense white fumes with a glass rod dipped in conc.HCl.	NH_4^+ may be present.
CONFIRMATORY TEST FOR AMMONIUM (NH_4^+) RADICAL:		
Salt solution in water + few drops of Nessler's reagent.	Brown precipitate is formed.	NH_4^+ is confirmed.
$\underset{\text{Nessler's reagent}}{2K_2[HgI_4]} + NH_3 + 3KOH \rightarrow \underset{\text{brown ppt.}}{H_2N.HgO.HgI \downarrow} + 7KI + 2H_2O$		

RESULT:

The given salt contains chloride (Cl^-) as acid radical and ammonium (NH_4^+) as basic radical.	
Hence the given salt is NH_4Cl – Ammonium chloride.	

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 3

A] PRELIMINARY TEST

4. Nature of the salt: Crystalline
5. Colour of the salt: Colourless
6. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H_2SO_4 in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings. Heated strongly.	No reaction	

DETECTION OF GROUP III ACID RADICAL(SO_4^{2-}):

Clear solution of the salt in dil.HCl+ Barium chloride solution.	A white precipitate is formed. It is insoluble in excess of dil.HCl	III group acid radical is present Sulphate is present

$SO_4^{-2} + BaCl_2 \rightarrow \underset{\text{white ppt.}}{BaSO_4} \downarrow + 2Cl^-$

CONFIRMATORY TEST FOR SULPHATE (SO_4^{-2})

Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution.	A white precipitate soluble in ammonium acetate solution	Sulphate is confirmed.

$SO_4^{-2} + Pb(CH_3COO)_2 \rightarrow \underset{\text{white ppt.}}{PbSO_4} \downarrow + 2CH_3COO^-$

C] ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.

PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	A gelatinous white precipitate.	III group basic radical present. Al^{3+} may be present.
CONFIRMATORY TEST FOR ALUMINIUM(Al^{3+})		
a) Original solution + NaOH solution dropwise. Above solution is treated with NH_4Cl solid. b) **Lake test:** Original solution + blue litmus solution + NH_4OH solution in excess.	Gelatinous white precipitate soluble in excess of NaOH. White gelatinous precipitate reappears. Bluish white precipitate, floating like lake is formed.	Al^{3+} is confirmed.

a) $Al^{3+} + 3\ OH^- \rightarrow \underset{\text{white ppt.}}{Al(OH)_3\downarrow}$

$Al(OH)_3 + OH^- \rightarrow \underset{\text{soluble complex}}{Al(OH)_4^-}$

b) $Al^{3+}(aq) + 3NH_4OH \rightarrow \underset{\text{blue color adsorbs on this ppt.}}{Al(OH)_3\downarrow} + 3NH_4^+$

RESULT:

The given salt contains sulphate(SO_4^{-2}) as acid radical and aluminium (Al^{+3}) as basic radical.	
Hence the given salt is $Al_2(SO_4)_3$ – Aluminium sulphate.	

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 4

A] PRELIMINARY TEST

4. Nature of the salt: Crystalline
5. Colour of the salt: Colourless
6. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS: **DETECTION OF GROUP I ACID RADICALS**		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.
DETECTION OF GROUP II ACID RADICALS:		
Salt + conc. H_2SO_4 in a dry test tube.	Reddish brown fumes and the solution turns brown.	Group II acid radical is present. Br^- may be present.
CONFIRMATORY TEST FOR CHLORIDE (Br^-)		
1.Silver Nitrate test: Clear salt solution in dilute Nitric acid + few drops of silver nitrate solution.	A pale yellow precipitate partially soluble in excess of ammonium hydroxide.	Br^- is confirmed.
$Br^- + AgNO_3 \rightarrow NO_3^- + \underset{\text{pale yellow ppt.}}{AgBr\downarrow}$		
2.Orange globule test: Clear salt solution in water + few drops of carbon tetrachloride + Chlorine water, shaken well.	Orange brown globule separates out.	Bromide is confirmed.
$Br^- + Cl_2 \rightarrow Cl^- + Br_2$ Bromine being soluble in CCl_4 imparts orange colour to CCl_4 layer.		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	A pungent smell of ammonia which gives dense white fumes with a glass rod dipped in conc.HCl.	NH_4^+ may be present.
CONFIRMATORY TEST FOR AMMONIUM (NH_4^+) RADICAL:		
Salt solution in water + few drops of Nessler's reagent.	Brown precipitate is formed.	NH_4^+ is confirmed.
$\underset{\text{Nessler's reagent}}{2K_2[HgI_4]} + NH_3 + 3KOH \rightarrow \underset{\text{brown ppt.}}{H_2N.HgO.HgI\downarrow} + 7KI + 2H_2O$		

RESULT:

The given salt contains <u>chloride (Br^-)</u> as acid radical and <u>ammonium (NH_4^+)</u> as basic radical.
Hence the given salt is NH_4Br – Ammonium bromide.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 5

A] PRELIMINARY TEST

1. Nature of the salt: Amorphous
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
-	+	

B] ANALYSIS OF ACID RADICALS: **DETECTION OF GROUP I ACID RADICALS**		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. HCl	Brisk effervescence is observed. A colorless gas is liberated.	Group I acid radical is present. Carbonate (CO_3^{-2}) may be present.
CONFIRMATORY TEST FOR CO_3^{-2}:		
Salt + Dil. HCl The above gas is passed through lime water.	Brisk effervescence is observed. A colorless gas is liberated. The lime water turns milky.	Carbonate (CO_3^{-2}) is confirmed.
$CO_3^{-2} + HCl \rightarrow CO_2 \uparrow + H_2O + Cl^-$ $Ca(OH)_2 + CO_2 \rightarrow \underset{\text{(milkiness)}}{CaCO_3 \downarrow} + H_2O$		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.

<u>DETECTION OF GROUP III BASIC RADICALS</u>		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
<u>DETECTION OF GROUP IV BASIC RADICALS</u>		
Original Solution + NH_4Cl solid +NH_4OH solution in excess + H_2S	No precipitate.	IV group basic radical absent.
<u>DETECTION OF GROUP V BASIC RADICALS</u>		
Original solution + NH_4Cl solid +NH_4OH solution in excess+$(NH_4)_2CO_3$ solution.	No precipitate.	V group basic radical absent.
DETECTION OF GROUP VI BASIC RADICALS		
Original solution + NH_4Cl solid + NH_4OH solution + ammonium phosphate solution.	White crystalline precipitate	VI group basic radical present. Mg^{2+} is present.
<u>CONFIRMATORY TEST FOR MAGNESIUM (Mg^{+2})</u>		
Original solution + NaOH solution.	White precipitate insoluble in NaOH solution	Mg^{2+} is confirmed.
$Mg^{2+} + NH_4OH + (NH_4)_2HPO_4 \rightarrow Mg(NH_4)PO_4\downarrow + H_2O$ $Mg^{2+} + 2NaOH \rightarrow Mg(OH)_2\downarrow + 2Na^+$		

<u>RESULT:</u>

The given salt contains <u>carbonate(CO_3^{-2})</u> as acid radical and <u>magnesium (Mg^{+2})</u> as basic radical.	
Hence the given salt is $MgCO_3$ – Magnesium carbonate.	

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 6

A] PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS: DETECTION OF GROUP I ACID RADICALS		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.
DETECTION OF GROUP II ACID RADICALS:		
Salt + conc. H_2SO_4 in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings. Heated strongly.	No reaction	
DETECTION OF GROUP III ACID RADICAL(SO_4^{2-}):		
Clear solution of the salt in dil.HCl+ Barium chloride solution.	A white precipitate is formed. It is insoluble in excess of dil.HCl	III group acid radical is present Sulphate is present
$SO_4^{-2} + BaCl_2 \rightarrow \underset{\text{white ppt.}}{BaSO_4} \downarrow + 2Cl^-$		
CONFIRMATORY TEST FOR SULPHATE (SO_4^{-2})		
Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution.	A white precipitate soluble in ammonium acetate solution	Sulphate is confirmed.
$SO_4^{-2} + Pb(CH_3COO)_2 \rightarrow \underset{\text{white ppt.}}{PbSO_4} \downarrow + 2CH_3COO^-$		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION:		

The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original solution + NH_4Cl solid +NH_4OH solution in excess. Warm + H_2S	White precipitate.	IV group basic radical present. Zn^{+2} may be present.
CONFIRMATORY TEST FOR ZINC (Zn^{2+})		
Dissolve the white precipitate in dilute HCl. Dilute it with 2- 3 mL of water. Divide the solution into two parts. i) To one part of the solution add dilute NaOH solution drop by drop till it is excess. ii) To another part of solution add 1mL of potassium ferrocyanide. Shake well.	White precipitate is soluble in excess of NaOH. White precipitate reappears. White (or bluish/grayish white) precipitate.	Zn^{+2} confirmed.
a) $Zn^{2+} + 2NaOH \rightarrow Zn(OH)_2 \downarrow + 2Na^+$ $Zn(OH)_2 + 2NaOH \rightarrow Na_2ZnO_2 + 2H_2O$ b) $2Zn^{2+} + K_4[Fe(CN)_6] \rightarrow Zn_2[Fe(CN)_6] \downarrow + 4K^+$		

RESULT:

The given salt contains sulphate(SO_4^{-2}) as acid radical and zinc (Zn^{+2}) as basic radical.	
Hence the given salt is $ZnSO_4$ – zinc sulphate.	

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 7

A] PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H_2SO_4 in a dry test tube.	Colourless gas is evolved. It gives dense white fumes with glass rod dipped in NH_4OH.	Group II acid radical is present. Cl- may be present.

CONFIRMATORY TEST FOR CHLORIDE (Cl^-)

1.Silver Nitrate test: Clear salt solution in dilute Nitric acid + few drops of silver nitrate solution.	A curdy white precipitate soluble in excess of ammonium hydroxide	Chloride is confirmed.

$Cl^- + AgNO_3 \rightarrow NO_3^- + \underset{\text{white ppt.}}{AgCl\downarrow}$

$AgCl + 2NH_4OH \rightarrow \underset{\text{soluble complex}}{[Ag(NH_3)_2]Cl} + 2H_2O$

2.Chromyl Chloride test: A pinch of salt + potassium dichromate crystals + few drops of conc.H_2SO_4. Pass the above vapours into NaOH solution. To the yellow solution add acetic acid and lead acetate solution.	Reddish brown vapours of Chromyl Chloride are evolved. Yellow solution. Yellow precipitate.	Chloride is confirmed.

$2Cl^- + K_2Cr_2O_7 + 3H_2SO_4 \rightarrow SO_4^{-2} + K_2SO_4 + \underset{\text{chromyl chloride}}{2CrO_2Cl_2} + 3H_2O$

$CrO_2Cl_2 + 4NaOH \rightarrow Na_2CrO_4 + 2H_2O + 2NaCl$

$Na_2CrO_4 + Pb(CH_3COO)_2 \rightarrow \underset{\text{Yellow ppt.}}{PbCrO_4\downarrow} + 2CH_3COONa$

C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original Solution + NH_4Cl solid +NH_4OH solution in excess + H_2S	No precipitate.	IV group basic radical absent.
DETECTION OF GROUP V BASIC RADICALS		
Original solution + NH_4Cl solid +NH_4OH solution in excess+$(NH_4)_2CO_3$ solution.	A white precipitate.	V group basic radical present. Ba^{2+}, Sr^{2+}, or Ca^{2+} may be present.
CONFIRMATORY TEST FOR BARIUM (Ba^{2+})		
Original solution +acetic acid + potassium chromate solution.	Yellow precipitate.	Ba^{2+} is confirmed.
$Ba^{2+} + 2CH_3COO^- \rightarrow (CH_3COO)_2Ba$ $(CH_3COO)_2Ba + K_2CrO_4 \rightarrow BaCrO_4 \downarrow + CH_3COOK$		
Flame test: A pinch of salt +few drops conc.HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Apple green colour is imparted to the flame.	Ba^{2+} is confirmed.

RESULT:

The given salt contains chloride(Cl^-) as acid radical and barium (Ba^{+2}) as basic radical.
Hence the given salt is $BaCl_2$ – Barium chloride.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 8

A] PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H_2SO_4 in a dry test tube.	No reaction	II group acid radical is present.
Above solution + Copper turnings. Heated strongly.	Reddish brown fumes and the solution turns blue.	Nitrate (NO_3^-) may be present.

CONFIRMATORY TEST FOR NITRATE(NO_3^-):

Brown ring test: Salt solution in dil.H_2SO_4 +freshly prepared saturated solution of Mohr's salt + conc.H_2SO_4 added carefully along the sides of the test tube.	A brown ring is formed at the junction of two liquids.	Nitrate (NO_3^-) is confirmed.

$NO_3^- + H_2SO_4 \rightarrow HSO_4^- + HNO_3$

$2HNO_3 + 3H_2SO_4 + 6FeSO_4 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$

$FeSO_4 + NO + 5H_2O \rightarrow \underbrace{[Fe(H_2O)_5(NO)]SO_4}_{\text{brown ring}} + H_2O$

C] ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.

PREPARATION OF ORIGINAL SOLUTION:
The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.

DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original Solution + NH_4Cl solid +NH_4OH solution in excess + H_2S	No precipitate.	IV group basic radical absent.
DETECTION OF GROUP V BASIC RADICALS		
Original solution + NH_4Cl solid +NH_4OH solution in excess+$(NH_4)_2CO_3$ solution.	A white precipitate.	V group basic radical present. Ba^{2+}, Sr^{2+}, or Ca^{2+} may be present.
CONFIRMATORY TEST FOR BARIUM (Ba^{2+})		
Original solution +acetic acid + potassium chromate solution.	No precipitate.	Ba^{2+} is absent.
CONFIRMATORY TEST FOR STRONTIUM (Sr^{2+})		
Original solution + ammonium sulphate solution, warm.	White precipitate.	Sr^{2+} is confirmed.
$Sr^{2+} + (NH_4)_2SO_4 \rightarrow SrSO_4 \downarrow + 2NH_4^+$		
Flame test: A pinch of salt +few drops conc.HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Crimson red colour is imparted to the flame.	Sr^{2+} is confirmed.

RESULT:

The given salt contains nitrate(NO_3^-) as acid radical and strontium(Sr^{+2}) as basic radical.
Hence the given salt is $Sr(NO_3)_2$– Strontium nitrate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 9

A] PRELIMINARY TEST

1. Nature of the salt: Amorphous
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
-	+	

B] ANALYSIS OF ACID RADICALS: **DETECTION OF GROUP I ACID RADICALS**		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. H_2SO_4	Brisk effervescence is observed. A colorless gas is liberated.	Group I acid radical is present. Carbonate (CO_3^{-2}) may be present.
CONFIRMATORY TEST FOR CO_3^{-2}:		
Salt + Dil. H_2SO_4 The above gas is passed through lime water.	Brisk effervescence is observed. A colorless gas is liberated. The lime water turns milky.	Carbonate (CO_3^{-2}) is confirmed.
$CO_3^{-2} + HCl \rightarrow CO_2 \uparrow + H_2O + Cl^-$ $Ca(OH)_2 + CO_2 \rightarrow \underset{\text{(milkiness)}}{CaCO_3 \downarrow} + H_2O$		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.

DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original Solution + NH_4Cl solid +NH_4OH solution in excess + H_2S	No precipitate.	IV group basic radical absent.
DETECTION OF GROUP V BASIC RADICALS		
Original solution + NH_4Cl solid +NH_4OH solution in excess+$(NH_4)_2CO_3$ solution.	A white precipitate.	V group basic radical present. Ba^{2+}, Sr^{2+}, or Ca^{2+} may be present.
CONFIRMATORY TEST FOR BARIUM (Ba^{2+})		
Original solution +acetic acid + potassium chromate solution.	No precipitate.	Ba^{2+} is absent.
CONFIRMATORY TEST FOR STRONTIUM (Sr^{2+})		
Original solution + ammonium sulphate solution, warm.	No precipitate.	Sr^{2+} is absent.
CONFIRMATORY TEST FOR CALCIUM (Ca^{2+})		
Original solution + NH_4OH solution + Ammonium oxalate solution.	White precipitate	Ca^{2+} is confirmed
$Ca^{2+} + (NH_4)_2C_2O_4 \rightarrow CaC_2O_4 \downarrow + 2NH_4^+$		
Flame test: A pinch of salt +few drops conc.HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Brick red colour is imparted to the flame.	Ca^{2+} is confirmed.

RESULT:

The given salt contains carbonate(CO_3^{-2}) as acid radical and calcium (Ca^{+2}) as basic radical.
Hence the given salt is $CaCO_3$ – Calcium carbonate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 10

A] PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

EXPERIMENT	OBSERVATION	INFERENCE
Salt + conc. H_2SO_4 in a dry test tube.	No reaction	II group acid radical is present.
Above solution + Copper turnings. Heated strongly.	Reddish brown fumes and the solution turns blue.	Nitrate (NO_3^-) may be present.

CONFIRMATORY TEST FOR NITRATE(NO_3^-):

EXPERIMENT	OBSERVATION	INFERENCE
Brown ring test: Salt solution in dil.H_2SO_4 +freshly prepared saturated solution of Mohr's salt + conc.H_2SO_4 added carefully along the sides of the test tube.	A brown ring is formed at the junction of two liquids.	Nitrate (NO_3^-) is confirmed.

$NO_3^- + H_2SO_4 \rightarrow HSO_4^- + HNO_3$

$2HNO_3 + 3H_2SO_4 + 6FeSO_4 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$

$FeSO_4 + NO + 5H_2O \rightarrow \underbrace{[Fe(H_2O)_5(NO)]SO_4}_{\text{brown ring}} + H_2O$

C] ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)

EXPERIMENT	OBSERVATION	INFERENCE
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.

PREPARATION OF ORIGINAL SOLUTION:
The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.

DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess + H_2S	No precipitate.	IV group basic radical absent.
DETECTION OF GROUP V BASIC RADICALS		
Original solution + NH_4Cl solid + NH_4OH solution in excess + $(NH_4)_2CO_3$ solution.	A white precipitate.	V group basic radical present. Ba^{2+}, Sr^{2+}, or Ca^{2+} may be present.
CONFIRMATORY TEST FOR BARIUM (Ba^{2+})		
Original solution +acetic acid + potassium chromate solution.	Yellow precipitate.	Ba^{2+} is confirmed.
$Ba^{2+} + 2CH_3COO^- \rightarrow (CH_3COO)_2Ba$ $(CH_3COO)_2Ba + K_2CrO_4 \rightarrow BaCrO_4 \downarrow + CH_3COOK$		
Flame test: A pinch of salt +few drops conc.HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Apple green colour is imparted to the flame.	Ba^{2+} is confirmed.

RESULT:

The given salt contains nitrate(NO_3^-) as acid radical and barium(Ba^{+2}) as basic radical.
Hence the given salt is $Ba(NO_3)_2$ – Barium nitrate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 11

A] PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS: **DETECTION OF GROUP I ACID RADICALS**		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.
DETECTION OF GROUP II ACID RADICALS:		
Salt + conc. H_2SO_4 in a dry test tube.	Colourless gas is evolved. It gives dense white fumes with glass rod dipped in NH_4OH.	Group II acid radical is present. Cl- may be present.
CONFIRMATORY TEST FOR CHLORIDE (Cl^-)		
1.Silver Nitrate test: Clear salt solution in dilute Nitric acid + few drops of silver nitrate solution.	A curdy white precipitate soluble in excess of ammonium hydroxide	Chloride is confirmed.
$Cl^- + AgNO_3 \rightarrow NO_3^- + \underset{\text{white ppt.}}{AgCl\downarrow}$ $AgCl + 2NH_4OH \rightarrow \underset{\text{soluble complex}}{[Ag(NH_3)_2]Cl} + 2H_2O$		
2.Chromyl Chloride test: A pinch of salt + potassium dichromate crystals + few drops of conc.H_2SO_4. Pass the above vapours into NaOH solution. To the yellow solution add acetic acid and lead acetate solution.	Reddish brown vapours of Chromyl Chloride are evolved. Yellow solution. Yellow precipitate.	Chloride is confirmed.
$2Cl^- + K_2Cr_2O_7 + 3H_2SO_4 \rightarrow SO_4^{-2} + K_2SO_4 + \underset{\text{chromyl chloride}}{2CrO_2Cl_2} + 3H_2O$ $CrO_2Cl_2 + 4NaOH \rightarrow Na_2CrO_4 + 2H_2O + 2NaCl$ $Na_2CrO_4 + Pb(CH_3COO)_2 \rightarrow \underset{\text{Yellow ppt.}}{PbCrO_4\downarrow} + 2CH_3COONa$		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		

TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original Solution + NH_4Cl solid +NH_4OH solution in excess + H_2S	No precipitate.	IV group basic radical absent.
DETECTION OF GROUP V BASIC RADICALS		
Original solution + NH_4Cl solid +NH_4OH solution in excess +$(NH_4)_2CO_3$ solution.	A white precipitate.	V group basic radical present. Ba^{2+}, Sr^{2+}, or Ca^{2+} may be present.
CONFIRMATORY TEST FOR BARIUM (Ba^{2+})		
Original solution +acetic acid + potassium chromate solution.	No precipitate.	Ba^{2+} is absent
CONFIRMATORY TEST FOR STRONTIUM (Sr^{2+})		
Original solution + ammonium sulphate solution, warm.	White precipitate.	Sr^{2+} is confirmed.
$Sr^{2+} + (NH_4)_2SO_4 \rightarrow SrSO_4 \downarrow + 2NH_4^+$		
Flame test: A pinch of salt +few drops conc.HCl and made into a paste. The paste is held to the flame with the help of a platinum wire.	Crimson red colour is imparted to the flame.	Sr^{2+} is confirmed.

RESULT:

The given salt contains chloride(Cl^-) as acid radical and strontium (Sr^{+2}) as basic radical.
Hence the given salt is $SrCl_2$ – Strontium chloride.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 12

A] PRELIMINARY TEST

1. Nature of the salt: Amorphous
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
-	+	

B] ANALYSIS OF ACID RADICALS: **DETECTION OF GROUP I ACID RADICALS**		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. HCl	Brisk effervescence is observed. A colorless gas is liberated.	Group I acid radical is present. Carbonate (CO_3^{-2}) may be present.
CONFIRMATORY TEST FOR CO_3^{-2}:		
Salt + Dil. HCl The above gas is passed through lime water.	Brisk effervescence is observed. A colorless gas is liberated. The lime water turns milky.	Carbonate (CO_3^{-2}) is confirmed.
$CO_3^{-2} + HCl \rightarrow CO_2 \uparrow + H_2O + Cl^-$ $Ca(OH)_2 + CO_2 \rightarrow \underset{\text{(milkiness)}}{CaCO_3 \downarrow} + H_2O$		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.

DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original Solution + NH_4Cl solid +NH_4OH solution in excess + H_2S	White precipitate.	IV group basic radical present. Zn^{+2} may be present.
CONFIRMATORY TEST FOR ZINC (Zn^{2+})		
Dissolve the white precipitate in dilute HCl. Dilute it with 2- 3 mL of water. Divide the solution into two parts. i) To one part of the solution add dilute NaOH solution drop by drop till it is excess. ii) To another part of solution add 1mL of potassium ferrocyanide. Shake well.	White precipitate is soluble in excess of NaOH. White precipitate reappears. White (or bluish/grayish white) precipitate.	Zn^{+2} confirmed.
a) $Zn^{2+} + 2NaOH \rightarrow Zn(OH)_2 \downarrow + 2Na^+$ $Zn(OH)_2 + 2NaOH \rightarrow Na_2ZnO_2 + 2H_2O$ b) $2Zn^{2+} + K_4[Fe(CN)_6] \rightarrow Zn_2[Fe(CN)_6] \downarrow + 4K^+$		

RESULT:

The given salt contains carbonate(CO_3^{-2}) as acid radical and zinc (Zn^{+2}) as basic radical.	
Hence the given salt is $ZnCO_3$ – Zinc carbonate.	

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 13

A] PRELIMINARY TEST

4. Nature of the salt: Crystalline
5. Colour of the salt: Colourless
6. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS: **DETECTION OF GROUP I ACID RADICALS**		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.
DETECTION OF GROUP II ACID RADICALS:		
Salt + conc. H_2SO_4 in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings.Heated strongly.	No reaction	
DETECTION OF GROUP III ACID RADICAL(SO_4^{2-}):		
Clear solution of the salt in dil.HCl+ Barium chloride solution.	A white precipitate is formed. It is insoluble in excess of dil.HCl	III group acid radical is present Sulphate is present
$SO_4^{-2} + BaCl_2 \rightarrow \underset{\text{white ppt.}}{BaSO_4} \downarrow + 2Cl^-$		
CONFIRMATORY TEST FOR SULPHATE (SO_4^{-2})		
Lead acetate test: Clear solution of the salt in water + Acetic acid + Lead acetate solution.	A white precipitate soluble in ammonium acetate solution	Sulphate is confirmed.
$SO_4^{-2} + Pb(CH_3COO)_2 \rightarrow \underset{\text{white ppt.}}{PbSO_4} \downarrow + 2CH_3COO^-$		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION: The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		

Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original solution + NH_4Cl solid +NH_4OH solution in excess. Warm + H_2S	Buff precipitate.	IV group basic radical present. Mn^{+2} may be present.
CONFIRMATORY TEST FOR MANGANOUS (Mn^{2+})		
Original solution + NH_4Cl solid +NH_4OH solution in excess + H_2S Add dil.HCl and boil. Add sodium hydroxide solution in excess	Buff / flesh coloured ppt. Precipitate dissolves. White precipitate formed and turns brown on exposure to air.	Mn^{2+} is confirmed.
$Mn^{+2} + H_2S \rightarrow MnS + H^+$ $MnS + 2HCl \rightarrow Mn\,Cl_2 + H_2S$ $Mn^{2+} + 2NaOH \rightarrow \underset{\text{white ppt.}}{Mn(OH)_2} \downarrow + 2Na^+$ $Mn\,(OH)_2 + [O] \rightarrow MnO(OH)_2$ **Hydrated manganese dioxide (Brown colour)**		

RESULT:

The given salt contains sulphate(SO_4^{-2}) as acid radical and manganous (Mn^{+2}) as basic radical.
Hence the given salt is $MnSO_4$ – Manganous sulphate.

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 14

A] PRELIMINARY TEST

7. Nature of the salt: Crystalline
8. Colour of the salt: Colourless
9. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS:

DETECTION OF GROUP I ACID RADICALS

EXPERIMENT	OBSERVATION	INFERENCE
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.

DETECTION OF GROUP II ACID RADICALS:

Salt + conc. H_2SO_4 in a dry test tube.	No reaction	Group II acid radical is absent.
Above solution + Copper turnings. Heated strongly.	No reaction	

DETECTION OF GROUP III ACID RADICAL(SO_4^{2-}):

Clear solution of the salt in dil.HCl+ Barium chloride solution.	No precipitate	III group acid radical sulphate is absent

DETECTION OF GROUP III ACID RADICAL(PO_4^{3-}):

Salt solution in water + conc. HNO_3 + solution of ammonium molybdate. Heat to boil.	A canary yellow precipitate is formed.	III group acid radical phosphate is present and confirmed

$$Na_3PO_4 + 3HNO_3 \rightarrow H_3PO_4 + 3NaNO_3$$

$$H_3PO_4 + 12(NH_4)_2MoO_4 + 21HNO_3 \rightarrow (NH_4)_3[P\ (Mo_3O_{10})_4] + 21NH_4NO_3 + 12H_2O$$

C] ANALYSIS OF BASIC RADICALS (CATIONS):

TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)

A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.

PREPARATION OF ORIGINAL SOLUTION:

The given salt is taken in a test tube and it is dissolved in minimum amount of water.

DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	No precipitate.	I group basic radical absent.
DETECTION OF GROUP II BASIC RADICALS		
Original solution + dil. HCl + H_2S solution is added.	No precipitate.	II group basic radical absent.
DETECTION OF GROUP III BASIC RADICALS		
Original Solution + NH_4Cl solid + NH_4OH solution in excess.	No precipitate.	III group basic radical absent.
DETECTION OF GROUP IV BASIC RADICALS		
Original solution + NH_4Cl solid +NH_4OH solution in excess. Warm + H_2S	White precipitate.	IV group basic radical present. Zn^{+2} may be present.
CONFIRMATORY TEST FOR ZINC (Zn^{2+})		
Dissolve the white precipitate in dilute HCl. Dilute it with 2- 3 mL of water. Divide the solution into two parts. i) To one part of the solution add dilute NaOH solution drop by drop till it is excess. ii) To another part of solution add 1mL of potassium ferrocyanide. Shake well.	White precipitate is soluble in excess of NaOH. White precipitate reappears. White (or bluish/grayish white) precipitate.	Zn^{+2} confirmed.

c) $Zn^{2+} + 2NaOH \rightarrow Zn(OH)_2 \downarrow + 2Na^{+}$

$Zn(OH)_2 + 2NaOH \rightarrow Na_2ZnO_2 + 2H_2O$

d) $2Zn^{2+} + K_4[Fe(CN)_6] \rightarrow Zn_2[Fe(CN)_6] \downarrow + 4K^{+}$

RESULT:

The given salt contains phosphate(PO_4^{-3}) as acid radical and zinc (Zn^{+2}) as basic radical.	
Hence the given salt is $Zn(PO_4)_3$ – zinc phosphate.	

SYSTEMATIC QUALITATIVE ANALYSIS OF SIMPLE INORGANIC SALT – 15

A] PRELIMINARY TEST

1. Nature of the salt: Crystalline
2. Colour of the salt: Colourless
3. Solubility:

Water(Cold/Hot)	Dil. HCl	Dil. HNO_3
+		

B] ANALYSIS OF ACID RADICALS: **DETECTION OF GROUP I ACID RADICALS**		
EXPERIMENT	**OBSERVATION**	**INFERENCE**
Salt + Dil. HCl	No brisk effervescence.	Group I acid radical is absent.
DETECTION OF GROUP II ACID RADICALS:		
Salt + conc. H_2SO_4 in a dry test tube.	No reaction	II group acid radical is present. Nitrate (NO_3^-) may be present.
Above solution + Copper turnings. Heated strongly.	Reddish brown fumes and the solution turns blue.	
CONFIRMATORY TEST FOR NITRATE(NO_3^-):		
Brown ring test: Salt solution in dil.H_2SO_4+freshly prepared saturated solution of Mohr's salt + conc.H_2SO_4 added carefully along the sides of the test tube.	A brown ring is formed at the junction of two liquids.	Nitrate (NO_3^-) is confirmed.
$NO_3^- + H_2SO_4 \rightarrow HSO_4^- + HNO_3$ $2HNO_3 + 3H_2SO_4 + 6FeSO_4 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$ $FeSO_4 + NO + 5H_2O \rightarrow \underbrace{[Fe(H_2O)_5(NO)]SO_4}_{\text{brown ring}} + H_2O$		
C] ANALYSIS OF BASIC RADICALS (CATIONS):		
TEST FOR AMMONIUM (NH_4^+) RADICAL: (ZERO GROUP ANALYSIS)		
A pinch of the salt is heated with 5 drops of sodium hydroxide.	No pungent smell.	Zero group basic radical absent.
PREPARATION OF ORIGINAL SOLUTION:		

The given salt is taken in a test tube and it is dissolved in minimum amount of water.		
DETECTION OF GROUP I BASIC RADICALS		
Original Solution + dil.HCl in a test tube.	White precipitate	I group basic radical present. Pb^{+2} may be present.
<u>CONFIRMATORY TEST FOR Pb^{+2} RADICAL:</u>		
Dissolve the precipitate in hot water and divide the solution into two parts. i) Part 1 + KI solution Part 2 + potassium chromate solution.	Yellow precipitate is formed. Yellow precipitate is formed.	Pb^{+2} is confirmed.
i) $Pb^{+2} + 2KI \rightarrow \underset{\text{yellow ppt.}}{PbI_2 \downarrow} + 2K^+$ ii) $Pb^{+2} + K_2CrO_4 \rightarrow \underset{\textbf{yellow ppt.}}{PbCrO_4 \downarrow} + 2K^+$		

<u>RESULT:</u>

The given salt contains <u>nitrate(NO_3^-)</u> as acid radical and <u>lead(Pb^{+2})</u> as basic radical.
Hence the given salt is $Pb(NO_3)_2$ – Lead nitrate.